Lebesgue's Measure Problem and Zermelo's Axiom of Choice: The Mathematical Effects of a Philosophical Dispute[a]

GREGORY H. MOORE

Department of Mathematics
University of Toronto
Toronto, Canada M5S 1A1

INTRODUCTION

During the early years of the twentieth century there arose an intriguing historical contradiction. This was not a contradiction within mathematics *per se*, such as was furnished during the same period by Russell's paradox, Burali-Forti's paradox, and the paradox of the largest cardinal. Rather, it was a contradiction between the expressed *philosophy* of a group of mathematicians and the type of *mathematics* that they developed. The contradiction was so fundamental that, if these mathematicians had understood it properly, their philosophical beliefs would have forced them to reject their life's work. At the very least, they would have had to recast it in a form that would destroy its usefulness to the mainstream of twentieth-century mathematics.

In order to understand this contradiction and its effects on the subsequent development of mathematics, we must first consider how Georg Cantor's set theory influenced French analysis late in the nineteenth century. In 1870 Cantor began to publish his findings on trigonometric series and, in particular, on what are now called their sets of uniqueness (*i.e.*, sets on which a real function can be represented by a unique trigonometric series). Little by little, this research led him to introduce his infinite ordinal and cardinal numbers as well as various concepts in point-set topology. Yet his ideas had a negligible impact on French mathematics until 1883 when Gösta Mittag-Leffler, editor of the new Swedish journal *Acta Mathematica*, printed French translations of several of Cantor's articles.[b] At the same time Cantor published in *Acta* a new but related paper [1883k], written in French.

The introduction of Cantor's ideas into France was largely due to two brothers, Jules and Paul Tannery. In the *Bulletin des sciences mathématiques et astronomiques*, whose audience consisted primarily of mathematicians, Jules Tannery [1884] favorably reviewed both the French translations of

[a]This paper was presented at a meeting held jointly by the Section of History, Philosophy and Ethical Issues of Science and Technology and the Section of Mathematics of The New York Academy of Sciences on March 25, 1981.

[b]The articles were Cantor, 1871, 1872, 1874, 1878, 1879, 1880, 1882, 1883, 1883a, and the corresponding translations were Cantor, 1883b–j. For a detailed and insightful study of Cantor's life and work, see Dauben, 1979.

0077-8923/83/0412-0129 $01.75/2 © 1983, NYAS

Cantor's work and his new paper. Indeed, Jules Tannery described Cantor's researches as constituting "a theory which, undoubtedly, is neither finished nor perfect but which is interesting from a philosophical viewpoint and which, from a mathematical perspective, is intimately connected to recent discoveries in the theory of [complex] functions . . ." [1884, p. 162]. In particular, Jules Tannery did not object to Cantor's infinite ordinal numbers or to the corresponding infinite cardinal numbers, as many later mathematicians were to do. Meanwhile, Paul Tannery was stimulated by one of the translations [Cantor, 1883c] to attempt a proof of Cantor's Continuum Hypothesis, which stated that every uncountable set of real numbers has the same cardinality as the set of all real numbers [P. Tannery, 1884]. Unfortunately, this proof was erroneous. The following year he introduced Cantor's work to a philosophical audience (the readers of the *Revue philosophique de la France et de l'etranger*) by describing how Cantor had characterized the notion of a continuum and by characterizing Cantor himself as "a pure idealist as to principles and methods" [P. Tannery, 1885, p. 406]. On the other hand, this Tannery had some reservations about Cantor's infinite ordinal numbers [1885, p. 410]. The following year Jules Tannery published a textbook on the theory of functions, in which he mentioned Cantor's construction of the real numbers, his derived sets, and his discovery that the rational numbers are countable—"a result which appears somewhat paradoxical" [J. Tannery, 1886, pp. x, 43, 58]. Much later he took pleasure in pointing out how his review of 1884 and his textbook of 1886 had "contributed to making G. Cantor's ideas known in France . . ." [J. Tannery, 1901, p. 29].

Several years passed between the contributions of the Tannerys and the next occasion on which a French mathematician utilized Cantor's work. In 1892 Camille Jordan proposed to study the role played by an arbitrary set E in the plane when a real function $f(x,y)$ is integrated over E. With this in mind, he remarked that earlier researchers had presupposed that every such E has a unique measure (*étendue*) and that this measure is finitely additive. That is, if $E_1 \cup E_2 = E$, where E_1 and E_2 are disjoint, then the measure of E equals the sum of the measure of E_1 and the measure of E_2 [1892, p. 69]. After establishing that an arbitrary E has a unique inner Jordan measure and a unique outer Jordan measure, he defined a set in the plane to be Jordan-measurable if its inner and outer Jordan measures are equal. In order to study Jordan measure more deeply, he utilized a number of Cantor's topological concepts such as limit point, derived set, and closed set [Jordan, 1892, p. 72]. When the second edition of Jordan's *Cours d'analyse* appeared the following year, it included both a substantial discussion of these Cantorian concepts and Jordan's theory of measure [1893, pp. 18–30].

Beginning in 1895, a number of French philosophers who were not also mathematicians became interested in Cantor's work. Arthur Hannequin, a young Kantian philosopher, wrote a book criticizing the use of "atoms" in physics and mathematics. Influenced by the philosopher Benno Kerry [1885], Hannequin devoted some twenty-four pages to Cantor and described in some detail the notion of derived set, the infinite ordinals, the characterization of the continuum as perfect and connected, and the status of the Continuum Hypothesis. Although Hannequin was willing to concede the existence of the

least infinite ordinal ω, he found Cantor's definition of a continuum to be inadequate, particularly because of its inability to deal with metric considerations such as length: "Thus Cantor's researches have served only to render more obvious the ancient conflict between the continuum on the one hand and the notion of [real] number on the other" [Hannequin 1895, p. 69]. The following year Cantor obtained a more sympathetic hearing from the philosopher Louis Couturat, whose doctoral dissertation [1896] contained seventy-five pages devoted to "the researches, so original and so profound, of Georg Cantor on the theory of infinite sets" [Couturat, 1896, p. 118]. Indeed, Couturat regarded the Continuum Hypothesis as already demonstrated [1896, p. 655], a position that he would modify the following year when reviewing Hannequin's book [Couturat, 1897, p. 781]. It is not surprising, however, that most of the other philosophers who criticized Cantor, such as Georges Lechalas [1897, pp. 627–629] and François Evellin [1898, pp. 115–116], did so by rejecting the actual infinite in favor of the potential infinite.

Until the mid-1890s, the philosophical views of French mathematicians had little effect on their usage of Cantorian notions. Then there emerged three French mathematicians—Emile Borel, René Baire, and Henri Lebesgue—who vigorously applied Cantorian notions while maintaining a philosophical position that would seriously undermine the usefulness of these notions. In this fashion, as we shall see, the contradiction mentioned at the beginning of this section was able to take root.

BOREL, BAIRE, AND LEBESGUE

In 1893 Emile Borel graduated from the *Ecole Normale Supérieure* in Paris. The following year, while teaching at the University of Lille, he completed a doctoral dissertation on the analytic continuation of a complex function, and soon was offered a teaching position at his alma mater. Various measure-theoretic ideas, including the notion of a set of measure zero, were implicit in this thesis [1895]. However, he first developed these ideas in detail within his book *Leçons sur la théorie des fonctions* [1898], which was dedicated to Jules Tannery. The first half of the book was devoted to Cantorian set theory. Reflecting on this period more than a decade later, Borel wrote: "I must admit that, like many young mathematicians, I was at first seduced by the Cantorian theory. I do not regret this, for it is a subject which renders the mind singularly flexible" [1912, p. 2].

Borel's monograph developed those parts of set theory which he deemed useful in complex analysis, in particular, elementary cardinal arithmetic and certain topological notions. Although he expressed some reservations about cardinal numbers greater than c (the power of the set of all real numbers), he accepted various theorems of Cantor's, such as the following:

> If a family A of sets has power c and if every set in A has power c, then the union of A also has power c [Borel, 1898, pp. 16–17]. **(1)**

If Borel had carefully examined his proof of this theorem, he would have found that it depended on the arbitrary choice, for each set B in A, of a one–one function from B onto c. The use of such arbitrary choices, which was becoming widespread late in the nineteenth century, raised few doubts at the time.[c] Yet such choices concealed a potent new mathematical assumption that was not formulated explicitly until 1904: the Axiom of Choice. This axiom states that if A is a family of non-empty sets, then there exists a function f such that $f(B) \in B$ for every set B in A. In more psychological terms, such a function f "chooses" an element from each set B in A. What remained unclear was how to specify such a function uniquely for a given family A.

It was this fact which made Borel an opponent of the Axiom of Choice when it was first stated explicitly in 1904. Indeed, Borel held definite philosophical views on the subject of what kind of mathematics was legitimate. In his opinion, mathematical objects had to be determined by a *rule,* that is, by a construction that uniquely specifies the object constructed. Yet in his monograph, unbeknownst to himself, Borel had repeatedly violated his own philosophical dictum by implicitly using the Axiom of Choice at several junctures, including theorem (1) above.

Eventually, although this was not apparent at first, the Axiom of Choice would greatly affect the theory of measure that Borel developed in his monograph. This theory of measure, he stressed, originated from quite different problems than had Jordan's theory. Restricting himself to the real interval [0,1], Borel wrote:

> When a set consists of all the points in a denumerable infinity of disjoint intervals having total length s, we say that the set *has measure* s. When two disjoint sets have measures s and s', their union has measure $s + s'$. . . . More generally, if we have a denumerable infinity of disjoint sets having respectively the measures $s_1, s_2, \ldots, s_n, \ldots$, their union . . . has measure
>
> $$s_1 + s_2 + \cdots + s_n + \ldots$$
>
> [countable additivity]. If a set E has measure s and contains every point in a set E' whose measure is s', *the set* $E - E'$ . . . *will be said to have measure* $s - s'$ . . . . *The sets that can be defined by the preceding definitions will be said to be measurable sets* . . . [1898, pp. 47–48].

In this fashion Borel introduced a class of measurable sets, which, following later usage, we shall call the Borel sets. At the time he did not develop their theory much further, except to prove that every closed set is a Borel set and that every countable set, as well as some uncountable sets, has measure zero.[d] It could easily be shown, he insisted in a note at the end of his book, that the

[c] At the time such choices raised doubts only in the mind of Giuseppe Peano [1890, p. 210] at Turin, and subsequently in the minds of two of his colleagues, Rodolfo Bettazzi [1896] and Beppo Levi [1902]. For a discussion of this matter, see Moore, 1982, pp. 26–29 and 76–82.

[d] In the present context a set is said to be finite if it is empty or has the same cardinality as the set $\{1, 2, \ldots, n\}$ for some natural number n; to be denumerable if it has the same cardinality as the set of all natural numbers; to be countable if it is either finite or denumerable; and to be uncountable if it is not countable.

following proposition is true:

> The set $\mathcal{B}$ of all Borel sets has the power c of the continuum [1898, p. 110]. **(2)**

Much later, however, mathematicians discovered that if the Axiom of Choice were false, it could happen that the set $\mathcal{B}$ has a power greater than c, that *every* set of real numbers is a Borel set, and that the hierarchy of the Borel sets collapses to the first three levels beyond the closed sets.[e]

The admiring but somewhat ambivalent attitude that Borel held toward Cantor can be seen in two articles which Borel wrote for the *Revue philosophique*. The first of these, published in October 1899, was stimulated by François Evellin's article [1898] in the same journal. Evellin had objected not only to Cantor's sequence of infinite cardinals of ever higher power but even to his use of the actual infinite at all. He wrote: "If, in order to define the irrational numbers, one introduces the notion of *set* in Cantor's sense, then one acts in a futile and dangerous manner . . ." [Evellin, 1898, p. 116]. In fact, Evellin asserted, the notion of the actual infinite is betrayed by an internal contradiction. Borel, on the other hand, approved of Cantor's notion of denumerable set (and hence of the actual infinite) but held certain reservations about the transfinite (*i.e.*, uncountable cardinals and ordinals):

> When, some twenty years ago, G. Cantor published his ideas on the enumeration of numbers larger than the infinite, the mathematical public greeted them mistrustfully. Nevertheless, many analysts were struck by the beauty of these speculations and did not let themselves be stopped by the form, intentionally paradoxical in places, of Cantor's exposition. . . . His new ideas have now . . . become classic, both for the mathematicians and for the philosophers [Borel, 1899, p. 383].

The remainder of this article was devoted to discussing du Bois-Reymond's theorem that, given any sequence of increasing real functions, there is a real function that increases still faster. By way of conclusion, Borel offered some reflections on the significance of that theorem for Cantor's work:

> One may . . . ask, quite simply, whether we can have a conception of the uncountable sequence of ever larger increasing functions. The difficulty in acquiring this conception comes from the fact that any *determinate* process of forming this sequence, since it is expressed by means of a finite number of words, leads only to a denumerable sequence. But, on the other hand, the theorem of Paul du Bois-Reymond tells us how to exceed this denumerable sequence once it is formed. Here is an antinomy which analysis appears incapable of resolving. It seems clear, however, that the subsequent development of mathematics will necessarily lead to a choice—to accept or to proscribe the *transfinite* (or uncountable infinity) [Borel, 1899, pp. 389–390].

In a second philosophical article, published the following year, Borel

[e]See Feferman and Levy, 1963, and the discussion in Moore, 1982, pp. 102–103. In particular, there is a model of Zermelo-Fraenkel set theory in which every set of real numbers is a countable union of countable sets.

expressed his reservations about the uncountable even more directly: "It seems to me that those who refuse to consider the notion of transfinite to be *as clear at present* as that of the indefinite [*i.e.*, the denumerable] are in the right" [1900, p. 382]. With the appearance in 1904 of Zermelo's Axiom of Choice, Borel's reservations would become much more pronounced.

Like his compatriot Borel, René Baire attended the *Ecole Normale Supérieure*. After graduating in 1895, Baire taught in various *lycées* and simultaneously completed his doctoral thesis, which was published in 1899. As Lebesgue later pointed out [1922, pp. 16–17], Baire "was the first to devote all his mathematical activity to the theory of real functions. . . ." Until the work of Baire, nineteenth-century analysts—particularly those in France—concerned themselves principally with the theory of complex functions. Baire's researches, which were influenced by the Italian analyst Vito Volterra, influenced Lebesgue in turn, and thereby brought about the flowering of the modern theory of real functions.

Baire shared with Borel a somewhat ambivalent attitude toward Cantorian set theory. In the conclusion to his thesis, which relied heavily on Cantor's results, Baire wrote:

> The theory of point-sets plays a very important role. . . . One can even say . . . that, from the perspective which we have adopted, every problem in the theory of [real] functions leads to certain problems in set theory; and it is to the extent that progress is made, or can be made, on these latter questions that it is possible to resolve more or less completely the given problem [1899, p. 121].

At the same time Baire refused to grant the existence of transfinite ordinals, which, however, he continued to use as superscripts of derived sets:

> On this subject I remark once and for all that we shall not be concerned with the difficulties that the abstract notion of *transfinite* [*ordinal*] *number* may involve, even though we may employ this expression in the rest of this work. In the present case, for example, the set P^{α} (where α is a definite number in the second number-class) represents something perfectly definite, independent of any abstract considerations relative to Cantor's symbols. Thus, in the usage that we make of the term *transfinite number* there is nothing more than a convenient language [1899, p. 36].

In his dissertation Baire introduced a hierarchy of classes of real functions. The essential feature of his hierarchy was that it treated discontinuous functions as the iterated limit of sequences of continuous functions. More precisely, he defined Baire class zero to be the set of all continuous real functions; for any other countable ordinal α, Baire class α was defined to be the set of all those real functions f, not in any previous class, such that

$$f(x) = \lim_{n \to \infty} f_n(x)$$

for every real x, where $f_1, f_2, \ldots$ was some sequence of real functions in previous classes. Thus, in particular, Baire class one was the set of all real functions that are the pointwise limit of a sequence of continuous functions. As a result, Cantor's ordinal numbers were essential to Baire's approach.

Moreover, by way of a framework for his classification, he explicitly accepted Dirichlet's general definition of a real function as any correspondence $x \to f(x)$ between real numbers. "In this definition," Baire observed, "it is beside the point to ask how the correspondence can be determined effectively or whether it is even possible to determine it effectively" [1899, p. 1]. This easy tolerance of non-constructive methods disappeared from Baire's work in the aftermath of Zermelo's introduction of the Axiom of Choice.

In order to derive results about his classification of real functions, Baire indirectly relied on the Axiom of Choice several times. One of the propositions that served as a conduit for such indirect uses of this axiom was the Countable Union Theorem, due to Cantor: The union of a countable family of countable sets is countable [1878, p. 243; 1883c, p. 313]. This theorem proved useful in establishing the basic properties of sets of first and second category, introduced by Baire the year before he published his thesis. Baire defined a set A of real numbers to be of first category if A is the union of countably many nowhere dense sets; otherwise, A was said to be of second category [1898]. Baire's thesis relied on the Countable Union Theorem to obtain the following elementary result:

> The union of countably many sets, each of first category, is also of first category [1899, p. 65]. (3)

Furthermore, he used the Countable Union Theorem to establish that no new functions would be added to his classification by extending it beyond the countable ordinals [1899, p. 70]. Finally, he relied implicitly on the Axiom to show that this classification did not exhaust all the real functions, since the set of functions in classification had cardinality c while the set of all real functions had a higher cardinality [1898, p. 1623]. If the Axiom of Choice were false, then all of Baire's results mentioned here could be false as well.

It is intriguing that Baire, who largely shared Borel's constructivist philosophy of mathematics, had a premonition that non-constructive methods might affect his own research. At first, Baire centered his investigations on Baire classes one and two. On 25 October 1898 he wrote to Volterra, whom he had visited in Italy not long before, in order to express his uncertainty about the existence of Baire class three and higher classes. Furthermore, Baire wondered whether it was possible to define a real function escaping from his classification:

> We would know how to obtain a function *beyond* all the [Baire] classes if we knew how to partition the continuum [of real numbers] into two sets, each of second category on every interval. We would let $f = 0$ for the first set and $f = 1$ for the second. I have not succeeded in *defining* such a partition, and I have come to wonder whether it is even *possible* to *define* such a partition. Clearly, it is a delicate matter not only to answer but even to pose such a question. The meaning of the phrase "to define a set" would have to be made more precise. Yet I am convinced that the act of imposing on a set [of real numbers] the condition of being definable must restrict the notion of set to a considerable degree. . . . I would not be surprised if it were impossible, with the means that we have

> available, to define a property partitioning the real numbers into two sets A and B such that both sets are of second category on every interval. We would have a proposition like the following, which is rather unexpected: If a set A and its complement B can be defined, then there exists a closed *nowhere dense* set P such that either A or B is of first category on any interval not intersecting P. . . . All of this is closely related to Borel's reflections in the notes at the end of his book on functions [1898]. . . . May we not hope to learn in this way how far we are permitted to use the notion of *arbitrary real function?* [Baire, quoted in Dugac, 1976, pp. 347–348]

In a sense, Baire's qualms were justified. The existence of such a partition as described in this quotation would imply the existence of a set of real numbers lacking what was later called the Baire property. By definition, a set A of real numbers has the Baire property if there is no interval such that A and its complement both intersect the interval in sets that are of second category. Around 1900, Baire pointed out to Lebesgue that it would be very interesting to define a set of real numbers that lacked the Baire property and hence to define a real function escaping from Baire's classification [Lebesgue, 1905, p. 186]. As later became clear, however, the existence of a set of real numbers lacking the Baire property, of a real function not in Baire's classification, and even of a real function not belonging to Baire class zero, one, or two, depend on the Axiom of Choice.

As Borel and Baire had done earlier, Henri Lebesgue attended the *Ecole Normale Supérieure.* After graduating in 1897, Lebesgue published a number of short papers, one of which generalized some of Baire's results to functions of several real variables [Lebesgue, 1899]. Shortly afterward, he made his first contribution to measure theory and gave the first version of his Measure Problem:

> The Measure Problem for plane regions bounded by simple closed curves can be posed in the following manner . . . : To associate with each such region a number, to be called *area,* such that congruent regions have equal areas and such that the region formed from the union of finitely or infinitely many regions, which have part of their boundary in common and which do not overlap, has as its area the sum of the areas of the component regions . . . [1899a, p. 870].

Here, in effect, Lebesgue adopted Borel's proposal that a measure should be countably additive.

Lebesgue's doctoral dissertation [1902] elaborated his new theory of measure and integration.[f] There he presented his notion of measurable set, which generalized the Borel sets: If A is a bounded set of real numbers in some fixed interval I of length k, the outer measure of A is defined to be the least upper bound of the sum of the lengths of countably many intervals covering A; the inner measure of A is defined to be $k - b$, where b is the outer measure of $I - A$. If the outer measure of A and the inner measure of A are equal, then A is said to be measurable [1902, pp. 237–238].

It was specifically in order to solve what he called the Measure Problem

[f]For an incisive historical treatment of Lebesgue's theory and its origins, see Hawkins, 1975.

that Lebesgue introduced his measurable sets. He posed this problem in a form that was quite general for his time:

> We propose to assign to each set [in n-dimensional space] a non-negative [real] number, which we shall call its measure, satisfying the following conditions:
>
> (i) There is a set whose measure is not zero.
>
> (ii) Congruent sets have equal measure.
>
> (iii) The measure of the sum [union] of a finite or denumerable infinity of disjoint sets is the sum of the measures of these sets.
>
> We shall resolve this *Measure Problem* only for the sets that we call measurable.[g]

To show that the measurable sets of real numbers are countably additive (condition (iii)), he first had to establish that the union of denumerably many disjoint measurable sets is measurable. He began by supposing that $A_1, A_2, \ldots$ is a sequence of disjoint measurable sets included in some real interval [a, b]. Next, he enclosed each set A_i in the union of some countable family $\mathcal{F}_i$ of open intervals, and each set [a, b] $- A_i$ in the union of another countable family $\mathcal{G}_i$ of open intervals, in such a way that each set $E_i = (\bigcup \mathcal{F}_i) \cap (\bigcup \mathcal{G}_i)$ had a total length d_i; here the families were chosen so that the sum of the d_i was an arbitrarily small positive number [1902, p. 239]. Nevertheless, Lebesgue provided no rule for choosing the families $\mathcal{F}_i$ and $\mathcal{G}_i$ for each i. Nor could he have done so, thereby avoiding arbitrary choices, since the Axiom of Choice is necessary to ensure the countable additivity of Lebesgue measure. More than a decade later, thanks to the researches of Wacław Sierpiński [1916; 1918], Lebesgue was faced with the dilemma of either accepting a limited form of the Axiom of Choice or else restricting his theory of measure and integration in an essential way. For, if the Axiom of Choice were false, it could happen that Lebesgue measure is not countably additive.[h]

From the additivity of his measure, Lebesgue concluded that every Borel set is measurable. Since, Lebesgue argued, there are only as many Borel sets as there are real numbers, and since there is a perfect nowhere dense set of measure zero, say A, then there is a measurable set that is not a Borel set. For the subsets of A are all measurable, being of measure zero, and the set of these subsets has a power greater than that of the real numbers, since A is perfect [Lebesgue, 1902, pp. 240–241]. Nevertheless, if the Axiom of Choice were false, then there could be just as many Borel sets as measurable sets, but there could also be a Borel set that is non-measurable.

Lebesgue also asserted that every real function in Baire's classification is a measurable function, where a real function f is said to be measurable if $\{x: a < f(x) < b\}$ is a measurable set for every real a and b. Indeed, Lebesgue showed that the sum of two measurable functions is measurable, and similarly for the product. However, when he proved that the pointwise limit f of a sequence f_i of measurable functions is measurable (provided that f is

[g] Lebesgue, 1902, p. 236. In his book on integration [1904, p. 103], he replaced condition (i) by the equivalent requirement that the n-dimensional unit cube have measure one.

[h] This is the case in the Feferman-Levy model mentioned in footnote *e*.

bounded), he used the proposition that the intersection of a countable family of measurable sets is measurable, and so relied indirectly on the countable additivity of Lebesgue measure. Hence his proof, whose immediate consequence was that every function in Baire's classification is measurable [Lebesgue, 1902, p. 257], used the Axiom of Choice in an essential way.

The measurable sets might not solve the Measure Problem, Lebesgue observed, since there might exist a bounded set of real numbers that was non-measurable. Moreover, he entertained the possibility that the Measure Problem might have a solution even if there existed a non-measurable set: "It has not been shown in any way that the Measure Problem is unsolvable for those sets (if any exist) whose outer and inner measures are unequal" [1902, p. 239]. In his book on the theory of integration Lebesgue maintained this viewpoint [1904, p. 106]. Furthermore, in an article published in 1905 but written early the previous year, he attempted to define a specific measurable set that is not a Borel set (not realizing that arbitrary choices had crept into his proof), and concluded by noting: "But the much more interesting question—can one specify a non-measurable set?—remains unanswered" [1905, p. 216]. The existence of such a non-measurable set turned out to be intimately related to Zermelo's Axiom of Choice.

ERNST ZERMELO AND THE WELL-ORDERING PROBLEM

In 1883 Cantor introduced the notion of a well-ordered set, that is, a linearly ordered set such that every non-empty subset has a least element.[f] Impressed by the properties of well-ordered sets and especially by the alephs (the cardinal numbers of infinite well-ordered sets), he asserted as a "law of thought" the proposition that every set can be well-ordered. By 1895 he had come to believe that this proposition required a proof. Not long afterward, he found an argument for this proposition and sent it to David Hilbert in 1896 or 1897, then to Richard Dedekind in 1899. However, Cantor would not permit this argument to be published, even though Philip Jourdain, who had found a similar argument [1904], asked permission to do so [Grattan-Guinness 1971, p. 115].

In August 1904 Julius König, a Hungarian mathematician, gave a lecture to the International Congress of Mathematicians at Heidelberg and purported to show that the set of all real numbers cannot be well-ordered. Cantor, who was present at König's lecture, could not find the gap in the argument. By the following day, however, Zermelo had isolated this gap [Kowalewski, 1950, p. 202].

Zermelo set to work in order to establish that Cantor's claim was correct: every set can be well-ordered. A month after König's lecture, Zermelo succeeded [1904]. He based his demonstration of this Well-Ordering Theo-

[f]Cantor, 1883a, p. 550. Strictly speaking, Cantor did not state this definition at the time, but one that was equivalent to it and considerably more complicated.

rem on a new postulate which soon came to be known as Zermelo's Axiom, or the Axiom of Choice. The proof gave rise to a controversy that, over the next four years, involved mathematicians in England, France, Germany, Holland, Hungary, Italy, and the United States [see Moore, 1978 and 1982].

This controversy entered France primarily via Hilbert, then an editor of *Mathematische Annalen,* to whom Zermelo had first sent his proof. On behalf of the *Annalen,* Hilbert asked Borel to prepare a reply to Zermelo's proof. Completed on 1 December 1904, Borel's brief article rejected the proof. What Zermelo had actually shown, Borel noted, was that the following two problems are equivalent for a given set M:

(A) to well-order M;
(B) to choose an element from every non-empty subset of M.

What Zermelo had *not* done, Borel insisted, was to prove that the equivalence of (A) and (B) provides a general solution for problem (A). To resolve (B) required "a means, at least a theoretical one, for determining a distinguished element m' from an arbitrary subset M'" of M [1905, p. 194]. In particular, Zermelo had supplied no such means for arbitrary sets of real numbers and hence had not shown that the set of all real numbers can be well-ordered. Borel closed his article by quoting part of a letter that he had received from Baire on the same subject. Baire doubted that it would ever be possible to well-order the set of all real numbers, since he considered this set and Cantor's infinite ordinals to be mere potentialities.

Borel's article stimulated a lively exchange of letters among those French mathematicians who at the time were applying Cantorian notions to other parts of mathematics. Writing to Borel, the analyst Jacques Hadamard disagreed with the conclusions of Borel's article. Hadamard accepted Zermelo's proof of the Well-Ordering Theorem because the proof used simultaneous choices, rather than successive choices depending on those made previously. Borel had claimed that "any argument which supposes an *arbitrary choice* to be made uncountably many times . . . [is] outside the domain of mathematics" [1905, p. 195]. Hadamard, on the other hand, found uncountably many arbitrary choices to be quite as acceptable as countably many. For Hadamard, the essential distinction was not one of cardinality, but the difference between showing that there exists a function with a given property and that a function with this property can be specified uniquely [Baire *et alii,* 1905, pp. 261–263].

Borel forwarded Hadamard's letter to Baire, who elaborated on his own previous letter. On the whole, Baire agreed with Borel, but considered himself to be more radical when he regarded every infinite set as merely a potential infinity established by convention. For a given infinite set A, Baire insisted, it was false to regard every subset of A as given. Consequently it made no sense to assume, as Zermelo had done, that an element was chosen from every non-empty subset of A. Zermelo's Axiom, Baire continued, was not contradictory but simply meaningless; and, in the final analysis, everything in mathematics must be reduced to the finite [Baire *et alii,* 1905, pp. 263–264].

Borel next solicited Lebesgue's opinion. Although at first he seemed to take the middle ground between Borel and Hadamard, Lebesgue gradually formulated a distinct but subtle position rejecting the Axiom of Choice. For Lebesgue, the principal question was whether one can prove the existence of a mathematical object without defining it. He admitted that it was a matter of convention how one answered this question, and conceded that sometimes he had published demonstrations in which he did not define a unique object of the type shown to exist. Nevertheless, he remained convinced that one could prove the existence of a given kind of mathematical object *only* by defining a unique particular instance of such an object. It was not enough to show that the class of all such objects was non-empty. In conclusion, Lebesgue differed from Borel by rejecting the Axiom of Choice not only for uncountable families of sets but for denumerable families as well [Baire *et alii*, 1905, pp. 264–269].

A NEGATIVE SOLUTION TO LEBESGUE'S MEASURE PROBLEM

In 1905, not long after this exchange of letters was published, an Italian mathematician applied the Axiom of Choice to Lebesgue's Measure Problem. Giuseppe Vitali, who had discovered the measurable sets independently of Lebesgue, familiarized himself with Lebesgue's book of 1904 on integration. By using the Axiom of Choice, Vitali established that there exists a subset of the real line which is not Lebesgue-measurable, *i.e.*, a non-measurable set. Vitali had a broader goal, however: to prove that Lebesgue's Measure Problem has no solution. In particular, he set out to show that there is no translation-invariant, countably additive, positive real measure defined on all bounded subsets of the real line and such that the unit interval has measure one.

Vitali began his proof by supposing that there was such a measure. For each real x, he let

$$A_x = \{x + b: b \text{ is rational}\}$$

and then designated H as the set of all such A_x. Selecting an element p in $A_x \cap (0, \frac{1}{2})$ for each distinct A_x in H, an act justified by the Axiom of Choice, he termed G_0 the set of all such p. Then he defined

$$G_q = \{p + q: p \in G_0\}$$

for each rational number q. Since the G_q are disjoint congruent sets, they all have the same measure. Moreover, each of $G_0, G_{1/2}, G_{1/3}, \ldots$ is a subset of the closed interval $[0, 1]$ and hence the sum of their measures is at most one. Yet by the countable additivity of the measure, each of them has measure zero. As a result, the measure of G_q is zero for every rational q, and so the measure of the real line (the union of all the G_q) is likewise zero—a contradiction. Consequently, Lebesgue's Measure Problem had no solution. Lebesgue's particular measure did not apply to every bounded set of real numbers, and so did not fulfill all the conditions that he had demanded of it. Vitali concluded

by noting that the set G_0 required some comment. This set certainly existed if the real numbers could be well-ordered. Those who did not grant the existence of such a well-ordering, he added, could conclude instead that there cannot be both a solution to the Measure Problem and a well-ordering of the real line [1905, pp. 3–5].

Two years later Lebesgue himself supplied an example of a set that is not Lebesgue-measurable, and simultaneously revealed how his philosophical presuppositions affected his attitude toward the existence of such a set. During 1905 he had written a monograph (not published until 1971) in which he adopted a distinction that had first been introduced some years earlier by Paul du Bois-Reymond [1887, p. 64], the distinction between "Idealists" and "Empiricists." According to Lebesgue, the Empiricists admitted the existence only of those real functions that could be uniquely defined, whereas the Idealists accepted the existence of real functions whose existence had been "proved" in some sense but which could not be uniquely defined [Lebesgue, 1971, pp. 37–39]. It is clear from the context that he regarded Zermelo and Hadamard as Idealists, while he considered Baire, Borel, and himself to be Empiricists. When he provided an example of a non-measurable set in 1907, and credited Vitali with an earlier example, Lebesgue insisted that its existence was established for Idealists, who accepted the Axiom of Choice, but not for Empiricists, who rejected this axiom [1907a, pp. 32–39]. Indeed, Lebesgue continued to doubt that a non-measurable set of real numbers would ever be defined uniquely [1907, pp. 202–203].

In 1914 Felix Hausdorff, a German-Jewish astronomer turned set theorist, investigated Lebesgue's Measure Problem in greater depth. Hausdorff, who accepted the Axiom of Choice completely, published an article [1914] in which he modified that problem. As a first step, he gave a demonstration somewhat different from Vitali's that the Measure Problem has no solution for the real line and hence lacks a solution for n-dimensional space whenever $n \geq 1$. He then inquired whether this problem would have a solution if the requirement (iii) of countable additivity were weakened to that of finite additivity:

(iv) $m(A \cup B) = m(A) + m(B)$ if A and B are disjoint.

While he was unable to resolve this weakened problem for $n = 1$ or $n = 2$, he succeeded in proving that there is no solution whenever $n \geq 3$.

Hausdorff's proof, which extended earlier arguments yielding a set that is not Lebesgue-measurable, showed that the existence of a solution for $n = 3$ would imply that one half of a sphere is congruent to one third of the same sphere. His strategy was to assume that there existed a measure satisfying (i), (ii), and (iv), and then to decompose a given sphere S into four sets A, B, C, D such that A, B, C, and $B \cup C$ were congruent to each other while D had measure zero. To obtain this decomposition, he let ϕ and ψ be rotations of S, through 180 and 120 degrees respectively, on distinct axes passing through the center of S. With ϕ, ψ, and ψ^2 treated as different formal factors, a group G of motions was generated. Each member of G had a unique representation either

as ϕ or as

$$\phi^{i_1}\psi^{j_1}\phi^{i_2}\psi^{j_2}\ldots\phi^{i_{n-1}}\psi^{j_{n-1}}\phi^{i_n}$$

for some n, where i_1 and i_n equaled zero or one, i_k equaled one if $1 < k < n$, and j_k equaled one or two. (What Hausdorff had obtained here, although he did not speak in these terms, was a free group with two generators.) The set D of those points fixed under members of G was denumerable, and was shown to have measure zero.

Finally, Hausdorff determined the sets A, B, and C. For each point x of $S - D$, he defined $P_x = \{g(x): g \in G\}$, and thus for every x and y in $S - D$, P_x and P_y were equal or disjoint. Using the Axiom of Choice to select a point from each distinct P_x, he called the resulting choice set M. Then $S - D$ was the union of the images of M under all g in G, and he partitioned $S - D$ into three sets A, B, and C by means of these images: For each g in G, exactly one of the two images $g''M$ and $(g\phi)''M$ was included in A and the other in $B \cup C$, while exactly one of the three images $g''M$, $(g\psi)''M$, and $(g\psi^2)''M$ was included in A, one in B, and one in C. It followed that A, B, C, and $B \cup C$ were all congruent to each other since $\psi''A = B$, $(\psi^2)''A = C$, and $\phi''A = B \cup C$. But then A had the measure of one half and of one third of the same sphere, a contradiction, completing the proof [1914, pp. 430–433].

Thanks to Borel, this result soon became known as Hausdorff's paradox. In the second edition [1914] of his book on set theory and complex functions, Borel included an exposition of Hausdorff's counterintuitive result, and concluded that the culprit was the Axiom of Choice:

> If, then, we designate by a, b, c, the probability that a point in S belongs to A, B, or C respectively and if we grant that the probability of a point belonging to a set E is not changed by a rotation about a diameter (this is what Lebesgue expresses by saying that two congruent sets have the same measure), one obtains the contradictory equalities: $a + b + c = 1$, $a = b$, $a = c$, $a = b + c$.
>
> The contradiction has its origin in the application . . . of Zermelo's *Axiom of Choice*. The set A is homogeneous on the sphere; but it is at the same time a half and a third of it. . . . The paradox results from the fact that *A is not defined*, in the logical and precise sense of the word *defined*. If one scorns precision and logic, one arrives at contradictions [Borel, 1914, pp. 255–256].

At last, Borel was convinced, his decade of opposition to the Axiom of Choice had been fully vindicated. Since this axiom generated a contradiction, its illogical character could not help but be obvious to mathematicians. In time, Borel's view came to be shared by a number of mathematicians who regarded Hausdorff's paradox (or its successor, the Banach-Tarski paradox) as sufficient reason to be wary of the Axiom of Choice.

LEBESGUE AND SIERPIŃSKI

It was Wacław Sierpiński who first realized how deeply the Axiom of Choice was entangled with Baire's classification of functions and with Lebesgue's theory of measure. On 4 December 1916, Emile Picard submitted to the Paris

Academy of Sciences a brief note by Sierpiński on the role of the Axiom of Choice in real analysis. There Sierpiński pointed out how the Denumerable Axiom of Choice (*i.e.*, the Axiom of Choice restricted to denumerable families of sets) was used in Baire's proof [1899, p. 69] that every function f in Baire class two is the iterated limit of some double sequence $f_{n,k}$ of continuous functions. Furthermore, Sierpiński observed that Lebesgue had implicitly used the Denumerable Axiom to show that Lebesgue measure is countably additive [Sierpiński, 1916, p. 690].

The following year the Paris Academy of Sciences printed another note by Sierpiński, who demonstrated that certain seemingly innocuous assumptions (having no obvious connection with the Axiom of Choice) implied the existence of a non-measurable set and hence depended on that axiom. In particular, he established that there is such a non-measurable set if the set of all real functions can be linearly ordered, or if the family of all denumerable sets of real numbers has the power c, or even if Baire class two has power c [1917].

Lebesgue [1918, p. 238] responded to Sierpiński's second article by claiming that Sierpiński had inadvertently used the Axiom of Choice to show that the existence of a non-measurable set follows from the proposition that the family of all denumerable sets of real numbers has power c. Nevertheless, Lebesgue was mistaken. As for Sierpiński's first article and, in particular, his assertion that the countable additivity of Lebesgue measure depends on the Denumerable Axiom, Lebesgue did not reply. In 1934 Charles de la Vallée-Poussin, another French analyst whose constructivistic views paralleled Lebesgue's, made a valiant but ultimately futile attempt to prove the countable additivity of Lebesgue measure without using the Axiom of Choice [de la Vallée-Poussin, 1934, p. 26].

In 1918 Sierpiński published a lengthy article analyzing the relationship between the Axiom of Choice and many propositions used in real analysis or set theory. Thereby he became the first to explain just how deeply the researches of Baire, Borel, and Lebesgue depended on that axiom. In order to understand Sierpiński's results, it will be helpful to return briefly to the article that Lebesgue wrote during 1904 and published the following year. In that article Lebesgue investigated a class of real functions which he called analytically representable. A real function f was said to be analytically representable if f is obtained from countably many variables and constants by the operations of addition, multiplication, and taking the limit of a sequence of functions. He argued that a real function f is in Baire's classification if and only if f is analytically representable [1905, p. 152]. After defining f to be a Borel function if and only if the set $\{x: a \leq f(x) \leq b\}$ is a Borel set for every real number a and b, he demonstrated that f is analytically representable if and only if f is a Borel function [1905, p. 168]. Furthermore, from the countable additivity of Lebesgue measure he deduced that every Borel set is measurable and that every function in Baire's classification is a measurable function. Finally, Lebesgue supplied an example of a real function that is measurable but not in Baire's classification, and of a measurable set that is not a Borel set [1905, pp. 212–216].

Sierpiński [1918] recognized that Lebesgue's results were thoroughly

entangled with the Axiom of Choice. In particular, Sierpiński observed, this axiom was used indirectly to ensure that every function in Baire's classification is analytically representable and hence measurable. On the other hand, he remarked that the Axiom of Choice appeared to be necessary to obtain the existence of a real function not in any Baire class or even of a function beyond Baire class three. In the absence of this axiom, he added, it was not known how to show that if $f_1(x), f_2(x), \ldots$ is a convergent sequence of functions in Baire's classification, then the function $f(x) = \lim_{n\to\infty} f_n(x)$ also belongs to Baire's classification, for such a demonstration relied on Cantor's Countable Union Theorem that the union of a countable family of countable sets is countable, a theorem that in turn depended on the Axiom of Choice [Sierpiński, 1918, pp. 133–137].

As a particularly important case of the Countable Union Theorem, Sierpiński focused on the following proposition:

> The set of all real numbers is not a countable union of countable sets. **(4)**

In fact, he continued, the countable additivity of Lebesgue measure implied **(4)**, and so did the apparently weaker proposition that the union of countably many sets having Lebesgue measure zero also has Lebesgue measure zero. Moreover, the Countable Union Theorem yielded a particular case of the Axiom of Choice, namely, the Denumerable Axiom restricted to families containing only countable sets [1918, pp. 113–114].

If **(4)** were false, then the hierarchy of Borel sets would be trivialized, as would Baire's classification of real functions. Instead of extending through all the countable ordinals (and even then including only a negligible portion of all sets of real numbers), the Borel hierarchy would be reduced to the first four levels. Every set of real numbers would be Δ_4^0 in modern notation, that is, a countable union of a countable intersection of a countable union of closed sets, as well as the complement of such a set. Every real function would be in some Baire class. Furthermore, every measurable set would be a Borel set, but Lebesgue measure would not be countably additive.

Not until 1963 was it shown that there is a model of Zermelo-Fraenkel set theory in which **(4)** is false. Using Paul Cohen's [1963] method of forcing, Solomon Feferman and Azriel Levy [1963] obtained such a model in which the Axiom of Choice is strongly violated. Thanks to recent research by Juris Steprāns, it is now known that in the Feferman-Levy model there is a Borel set that is non-measurable. In this way Sierpiński's intuition about the role of the Axiom of Choice in the researches of Baire, Borel, and Lebesgue was fully corroborated.

As late as 1938, Lebesgue actively continued to oppose the Axiom of Choice. In December of that year, a conference on the foundations of mathematics was held in Zurich. Among the participants were Lebesgue and Sierpiński, who gave consecutive lectures about the Axiom of Choice. Lebesgue congratulated Sierpiński and his students on their extensive researches on the axiom. All the same, Lebesgue remained a subtle adversary

of this assumption. When the controversy over the axiom began in 1905, he observed, the opposing sides could not make themselves understood, because they lacked a common logic; indeed, logic itself was at stake [Lebesgue, 1941, p. 116].

Yet Lebesgue knew quite well that the issues involved were not limited to the underlying logic. Despite the fact that during three decades of active use the Axiom of Choice had not led to a contradiction, there remained a sense of unease. For logic did not create confidence in a proposition, he insisted, unless one had already accepted the proposition as plausible. By way of conclusion, he argued that "in the studies on the foundations and methods of mathematics, there must be a large place for psychology and even for esthetics" [1941, p. 122]. Thereby he underlined the psychologistic perspective that he had shared with Baire and Borel at least since 1905.

What is surprising in this controversy is how little effort Baire, Borel, and Lebesgue appear to have made to understand precisely how the Axiom of Choice affected their discoveries. Lebesgue, in particular, seems never to have attempted seriously to grasp how essential this axiom was to the countable additivity of his measurable sets and hence to the countable additivity of the Lebesgue integral. Only Leonida Tonelli in Italy endeavored to develop a version of the Lebesgue integral without the Axiom of Choice and consequently without countable additivity. However, Tonelli's research [1921] attracted little interest.

GENERALIZING LEBESGUE'S MEASURE PROBLEM

Hausdorff's research of 1914 on the Measure Problem was continued by a young Polish mathematician, Stefan Banach, who had recently written his doctoral thesis [1922] at Lwow on linear operators and who was soon to develop a school of functional analysis there. Banach published in 1923 an article devoted to what he called Hausdorff's Extended Measure Problem:

> To find a non-negative function m, defined on every bounded subset A of n-dimensional Euclidean space, such that
> (1) $m(A)$ is positive for some A.
> (2) Congruent sets have equal measures.
> (3) $m(A \cup B) = m(A) + m(B)$ if A and B are disjoint.

Like the majority of Polish mathematicians, Banach accepted the Axiom of Choice. Consequently, he accepted as well Hausdorff's result that the Extended Measure Problem has no solution for $n \geq 3$.

All the same, the Extended Measure Problem remained open for the cases $n = 1$ and $n = 2$: the Euclidean line and plane. By means of the Well-Ordering Theorem Banach established that the Extended Measure Problem has a positive solution in these two cases [1923, p. 19]. Thus an additive, congruence-preserving measure could be defined on all subsets of the line, and another such measure on all subsets of the plane. Moreover, in each case this measure could be assigned so as to agree with Lebesgue measure for all

Lebesgue-measurable sets, or so as to assign the measure one to some set with Lebesgue measure zero. As a result, Banach obtained two types of integral, each of which was defined on all subsets of the line or of the plane. One of these extended the Lebesgue integral, whereas the other extended the Riemann integral but differed at times from the Lebesgue integral. Both of Banach's integrals were additive but, necessarily, neither was countably additive.[j]

Because of the disparity of the Extended Measure Problem between the cases $n = 1$ and $n = 2$ on the one hand and $n \geq 3$ on the other, Banach attempted to come to a deeper understanding of Hausdorff's proof for the latter case. During the same period Alfred Tarski, one of Sierpiński's former students at Warsaw, undertook some related research. Banach and Tarski wrote a joint paper showing, among other things, that it is impossible to partition a plane polygon P into a finite number of pieces and reassemble them to obtain a polygon strictly including P [1924]. By contrast, they established that in three dimensions a polyhedron—and indeed *every* bounded set with non-empty interior—can be partitioned into a finite number of pieces and reassembled to give any other bounded set with non-empty interior. Thus, in particular, a sphere of unit radius could be partitioned into a finite number of pieces and reassembled into two spheres, each of unit radius. This implausible result became known to later mathematicians as the Banach-Tarski paradox.[k]

As for the role of the Axiom of Choice in their two decomposition theorems, the first for polyhedra in space and the second for polygons in the plane, Banach and Tarski were very explicit. They gave no encouragement to opponents of this axiom:

> One does not know how to establish either of these two theorems without relying on the Axiom of Choice—neither the first, which seems paradoxical, nor the second, which completely agrees with one's intuition. Furthermore, by analyzing their respective proofs, one can verify that the Axiom of Choice plays a much less substantial role in the paradoxical theorem than it does in the theorem agreeing with intuition [1924, p. 245].

Eventually, however, A. P. Morse [1949] established that the non-existence of paradoxical decompositions in the line and on the plane could be proved *without* this axiom.

In 1929 John von Neumann shed light on the Banach-Tarski paradox and, at the same time, proposed a more far-reaching generalization of the Measure Problem. It might appear, von Neumann observed, that the existence of a

[j]Banach, 1923, pp. 23–33. When he returned to these questions in his well-known book *The Theory of Linear Operators*, he derived his solution to the Extended Measure Problem from the Hahn-Banach Theorem [Banach, 1932, pp. 29–33]. It should be noted that this theorem too requires the Axiom of Choice.

[k]According to a personal communication in March 1982 from Tarski to the author, Banach and Tarski independently discovered the Banach-Tarski paradox, and then they decided to write a joint paper on the subject [1924]; however, the more general result on three-dimensional bounded sets with non-empty interiors was developed by Tarski alone.

suitable measure in one or two dimensions—as contrasted with the non-existence of such a measure in three or more dimensions—revealed a fundamental dichotomy between the nature of the Euclidean line or plane and that of Euclidean space. As he hastened to point out, this was not true. The essential difference did not arise from the dimensionality of the space but from the underlying group of motions. In order to clarify the matter, he revised Hausdorff's Extended Measure Problem by letting M be any set, W be any subset of M, and G be any group of one-one mappings of M onto itself. Then he defined m to be an (M, W, G)-measure if m assigned a non-negative real number to each subset A of M and if m had the following three properties:

(a) $m(W) = 1$.
(b) If f is in G, then $m(A) = m(f''A)$, where $f''A$ is the image of A under f.
(c) $m(A \cup B) = m(A) + m(B)$ if A and B are disjoint.

Here he altered Hausdorff's problem in two essential respects, first by replacing n-dimensional Euclidean space with an arbitrary set, and second by considering an arbitrary group instead of the congruence-preserving group of Euclidean motions. After he stated a sufficient condition (which involved Abelian factor groups) for an (M, W, G)-measure to exist, he proved that this condition is satisfied by the group of Euclidean motions in n-dimensional space for $n = 1$ and $n = 2$, but not for $n \geq 3$. On the other hand, he demonstrated that if G has a free subgroup with two generators and without fixed points (as was the case for the group of Euclidean motions in n-dimensional space whenever $n \geq 3$), then an (M, M, G)-measure does not exist. Finally, he showed that an (M, M, G)-measure fails to exist for the plane too if the group G is taken to be the affine group.[1] In all of these results, the Axiom of Choice played an essential role [1929, p. 73–87].

Later the same year, the Measure Problem was generalized in a different direction by Banach, who regarded the problem as essentially set-theoretic rather than geometric or even group-theoretic. By means of his generalization, he placed the problem (which he called the Generalized Measure Problem) squarely in the abstract theory of measure. This problem asked for a function m that assigns a non-negative real number to each subset A of the closed real interval $[0, 1]$ and satisfies the following conditions:

(I) $m(A) \neq 0$ for some A.
(II) $m(A) = 0$ if A contains exactly one element.
(III) m is countably additive.

Thereby Banach dispensed with the geometric requirement that congruent sets have equal measure and replaced it by the requirement that a point have

[1]Von Neumann also showed that a certain plausible choice of G, which, however, was not measure-preserving, led to a paradoxical decomposition of any line segment. During the summer of 1982 Stanley Wagon succeeded in showing that for the proper choice of G there fails to exist a (M, M, G)-measure on any line segment M. Wagon's proof was substantially simplified by Jan Mycielski during the fall (personal communication from S. Wagon).

measure zero. Without this latter requirement there would have been a trivial solution, namely, $m(A) = 1$ if A contains a certain predetermined point and $m(A) = 0$ otherwise. By weakening the geometric requirement of preserving congruence to an essentially set-theoretic condition, Banach was able to inquire if countable additivity could be reinstated.

In 1929 Banach published a joint article with another of Sierpiński's former students, Kazimierz Kuratowski. After Banach posed the Generalized Measure Problem, he found that he could solve it negatively if he could demonstrate a certain combinatorial lemma on countable unions. Banach and Kuratowski independently arrived at a proof of this lemma, by means of the Continuum Hypothesis, and so they published their results in a joint paper [1929]. Consequently, no measure of the desired sort existed if the Continuum Hypothesis was true.

Soon Banach took cognizance of the fact that the Generalized Measure Problem did not really concern the interval [0, 1] in particular but rather any set with the cardinality c of the continuum. As a result, one could pose the same problem for sets of arbitrary infinite cardinality. Thus in 1930 Banach introduced the Generalized Measure Problem for Abstract Sets: to find, for a given set E of cardinality $\aleph_\alpha$, a function m that assigns a non-negative real number to each subset A of E such that conditions (I), (II), and (III) hold. By using transfinite induction on α and by assuming the Generalized Continuum Hypothesis ($2^{\aleph_\beta} = \aleph_{\beta+1}$ for all ordinals β), he was able to show that the cardinal of the smallest E to have such a measure was weakly inaccessible [1930]. That is, the cardinal of E was uncountable, not the successor of any cardinal, and not the sum of a smaller number of smaller cardinals.

Meanwhile, Tarski had weakened Banach's Generalized Measure Problem for Abstract Sets by replacing condition (III) of countable additivity with condition (iv) of finite additivity. By means of the Axiom of Choice, Tarski showed that every infinite set E has such a measure and, furthermore, that the measure can be restricted to take only the values zero and one [1930]. In effect, Tarski obtained his result by constructing a maximal ideal on the family of subsets of E.

As Tarski acknowledged at the end of his paper [1930, p. 50], Kuratowski had pointed out to him that his result also followed easily from the argument in an earlier article by Stanisław Ulam [1929]. At that time Ulam was a twenty-year-old student at Lwow who, as a result of his paper of 1929, had decided to become a mathematician [1976, p. 30]. Encouraged by Kuratowski, Ulam attempted to improve Banach's result. Without assuming the Generalized Continuum Hypothesis or even the Continuum Hypothesis, Ulam showed that the cardinal of the smallest set E satisfying the Generalized Measure Problem for Abstract Sets must be weakly inaccessible and that the Generalized Measure Problem has no solution if there is no weakly inaccessible cardinal less than or equal to c [1930, p. 141]. As in the results of Banach and Kuratowski, the Axiom of Choice was essential here.

In more recent terminology, the cardinal of a set E satisfying the Generalized Measure Problem for Abstract Sets is said to be a real-valued measurable cardinal, whereas such a cardinal whose measure takes only the

values zero and one is called a measurable cardinal. Thus Ulam established that the first measurable cardinal is strongly inaccessible (*i.e.*, weakly inaccessible and not the product of a smaller number of smaller sets) and that the first real-valued measurable cardinal is likewise strongly inaccessible if every weakly inaccessible cardinal is greater than c [1930, p. 150]. Independently, Tarski had shown that if a cardinal is non-measurable, then the cardinal of its power set is non-measurable [Ulam, 1930, p. 146].

Yet the question remained open as to whether there exists a measurable cardinal or even a real-valued measurable cardinal. Not until 1960 did Tarski (relying heavily on the work of his student William Hanf) establish that the first measurable cardinal κ, if it exists, must be extraordinarily large. In particular, there must be κ strongly inaccessible cardinals less than κ [Tarski, 1962].

CONCLUSION

In the absence of the Axiom of Choice, however, there were other possibilities. Indeed, it could then happen, as Thomas Jech demonstrated [1968], that aleph-one is a measurable cardinal—provided that there is a model of set theory with the Axiom of Choice in which there is a measurable cardinal. Moreover, Robert Solovay established that c may be a real-valued measurable cardinal since every set of real numbers can be Lebesgue-measurable [1965; 1970]. To obtain this result, Solovay had to postulate the existence of a strongly inaccessible cardinal—hardly an agreeable assumption to constructivists such as Baire, Borel, and Lebesgue.[m]

Thus Lebesgue's Measure Problem, which originated as a technical problem in mathematics, acquired philosophical ramifications—thanks to the Axiom of Choice. Subsequently, the philosophical problems existed side by side with new mathematical results. Lebesgue's philosophical scruples prevented him from pursuing fully the questions that his Measure Problem led others to raise. At the same time, his work reveals how a mathematician of the first rank may subtly fail to see how he is fundamentally violating his philosophical scruples in his own work. By contrast, Sierpiński and his Polish colleagues, without insisting that one accept the Axiom of Choice, explored its effects on the Measure Problem with clarity, depth, and consistency.

ACKNOWLEDGMENTS

A preliminary version of this paper was read at the University of Toronto on 22 March 1978. The present version was read at the New York Academy of Sciences on 25 March 1981, at Iowa State University on 9 February 1982, and

[m]The necessity of the inaccessible cardinal (provided that the Denumerable Axiom holds or at least that aleph-one is regular) is due to Saharon Shelah (1980). Solovay (1971) had also shown that, in set theory with the Axiom of Choice, the existence of a measurable cardinal is equiconsistent with the existence of a real-valued measurable cardinal.

at the University of Waterloo on 13 February 1982. I would like to thank Ivor Grattan-Guinness, Esther R. Phillips, and Stanley Wagon for their suggestions, both stylistic and substantive.

REFERENCES

BAIRE, R.
1898 Sur les fonctions discontinues qui se rattachent aux fonctions continues. Comptes Rendus Hebdomadaires des Séances de l'Académie des Sciences, Paris **129:** 1621–1623.
1899 Sur les fonctions de variables réelles. Annali di matematica pura ed applicata (3) **3:** 1–123.

BAIRE, R., *et alii*
1905 Cinq lettres sur la théorie des ensembles. Bulletin de la Société Mathématique de France **33:** 261–273; translated in Moore, 1982, pp. 311–320.

BANACH, S.
1922 Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales. Fundamenta Mathematicae **3:** 133–181.
1923 Sur le problème de la mesure. *ibid.* **4:** 7–33.
1930 Über additive Massfunktionen in abstrakten Mengen. *ibid.* **15:** 97–101.
1932 Théorie des opérations linéaires. Monografje Matematyczne, vol. 1 (Warsaw: Garasiński); reprinted by Chelsea, New York, 1963.

BANACH, S. & K. KURATOWSKI
1929 Sur une généralisation du problème de la mesure. Fundamenta Mathematicae **14:** 127–131.

BANACH, S. & A. TARSKI
1924 Sur la décomposition des ensembles de points en parties respectivement congruentes. Fundamenta Mathematicae **6:** 244–277.

BETTAZZI, R.
1896 Gruppi finiti ed infiniti de enti. Accademia delle Scienze di Torino, Classe di Scienze Fisiche, Matematiche, e Naturale (Atti) **31:** 506–512.

BOREL, E.
1895 Sur quelques points de la théorie des fonctions. Annales scientifiques de l'Ecole Normale Supérieure **12:** 9–55.
1898 Leçons sur la théorie des fonctions. Gauthier-Villars. Paris.
1899 A propos de 'l'infini nouveau.' Revue philosophique de la France et de l'étranger **48:** 383–390.
1900 L'antinomie du transfini. *ibid.* **49:** 378–383.
1905 Quelques remarques sur les principes de la théorie des ensembles. Mathematische Annalen **60:** 194–195.
1912 Notice sur les travaux scientifiques de M. Emile Borel. Gauthier-Villars. Paris.
1914 Second edition of Leçons sur la théorie des fonctions, 1898.

CANTOR, G.
1871 Über trigonometrische Reihen. Mathematische Annalen **4:** 139–143.
1872 Über die Ausdehnung eines Satzes aus der Theorie der trigonometrischen Reihen. *ibid.* **5:** 123–132.
1874 Über eine Eigenschaft des Inbegriffes aller reellen algebraischen Zahlen. Journal für die reine und angewandte Mathematik (Crelle) **77:** 258–262.
1878 Ein Beitrag zur Mannigfaltigkeitslehre. *ibid.* **84:** 242–258.

1879 Über unendliche lineare Punktmannichfaltigkeiten, I. Mathematische Annalen **15:** 1–7.
1880 Part II of [1879]. *ibid.* **17:** 355–358.
1882 Part III of [1879]. *ibid.* **20:** 113–121.
1883 Part IV of [1879]. *ibid.* **21:** 51–58.
1883a Part V of [1879]. *ibid.* **21:** 545–586.
1883b Sur une propriété du système de tous les nombres algébriques réels. Acta Mathematica **2:** 305–310 (French translation of [1874]).
1883c Une contribution à la théorie des ensembles. *ibid.*: 311–328 (French translation of [1878]).
1883d Sur les séries trigonométriques. *ibid.*: 329–335 (French translation of [1871]).
1883e Extension d'un théorème de la théorie des séries trigonométriques. *ibid.*:336–348 (French translation of [1872]).
1883f Sur les ensembles infinis et linéaires de points, I. *ibid.*: 349–356 (French translation of [1879]).
1883g Part II of [1883f]. *ibid.*: 357–360 (French translation of [1880]).
1883h Part III of [1883f]. *ibid.*: 361–371 (French translation of [1882]).
1883i Part IV of [1883f]. *ibid.*: 372–380 (French translation of [1883]).
1883j Fondements d'une théorie générale des ensembles. *ibid.*: 381–408 (French translation of [1883a]).
1883k Sur divers théorèmes de la théorie des ensembles de points situés dans un espace continu à *n* dimensions. Première communication. Extrait d'une lettre addressée à l'éditeur. *ibid.*: 409–414.

COHEN, P. J.
1963 The Independence of the Continuum Hypothesis, I. Proceedings of the National Academy of Sciences USA **50:** 1143–1148.

COUTURAT, L.
1896 De l'infini mathématique. Alcan. Paris.
1896a Review of Hannequin, 1895. Revue de Métaphysique et de Morale **4:** 778–797.
1897 Continuation of 1896a. *ibid.* **5:** 87–113, 220–247.

DAUBEN, J. W.
1979 Georg Cantor: His Mathematics and Philosophy of the Infinite. Harvard University Press. Cambridge, MA.

DE LA VALLÉE-POUSSIN, C.
1934 Intégrales de Lebesgue, fonctions d'ensembles, classes de Baire, second edition. Gauthier-Villars. Paris.

DUGAC, P.
1976 Notes et documents sur la vie et l'oeuvre de René Baire. Archive for History of Exact Sciences **15:** 297–383.

EVELLIN, F.
1898 L'infini nouveau. Revue philosophique de la France et de l'étranger **45:** 113–119.

FEFERMAN, S. & A. LEVY
1963 Independence Results in Set Theory by Cohen's Method, II. Notices of the American Mathematical Society **10:** 593.

GRATTAN-GUINNESS, I.
1971 The Correspondence between Georg Cantor and Philip Jourdain. Jahresbericht der Deutschen Mathematiker-Vereinigung **73:** 111–130.

HANNEQUIN, A.
1895 Essai critique sur l'hypothèse des atomes dans la science contemporaine. Masson. Paris.
HAUSDORFF, F.
1914 Bemerkung über den Inhalt von Punktmengen. Mathematische Annalen **75:** 428–433.
1914a Grundzüge der Mengenlehre. Veit. Leipzig.
HAWKINS, T.
1975 Lebesgue's Theory of Integration, second edition. Chelsea. New York.
JECH, T.
1968 ω_1 Can Be Measurable. Israel Journal of Mathematics **6:** 363–367.
JORDAN, C.
1892 Remarques sur les intégrales définies. Journal de mathématiques pures et appliquées (4) **8:** 69–99.
1893 Cours d'analyse de l'Ecole Polytechnique: Tome premier, Calcul différentiel, second edition. Gauthier-Villars. Paris.
JOURDAIN, P. E. B.
1904 On the Transfinite Cardinal Numbers of Well-Ordered Aggregates. Philosophical Magazine (6) **7:** 61–75.
KERRY, B.
1885 Über G. Cantor's Mannigfaltigkeitsuntersuchungen. Vierteljahrsschrift für wissenschaftliche Philosophie **9:** 191–232.
KOWALEWSKI, G.
1950 Bestand und Wandel. Oldenbourg. Munich.
LEBESGUE, H.
1899 Sur les fonctions de plusieurs variables. Comptes Rendus Hebdomadaires des Séances de l'Académie des Sciences, Paris **128:** 811–813.
1899a Sur la définition de l'aire d'une surface. *ibid.* **129:** 870–873.
1902 Intégrale, longueur, aire. Annali di matematica pura ed applicata (3) **7:** 231–359.
1904 Leçons sur l'intégration et la recherche des fonctions primitives. Gauthier-Villars. Paris.
1905 Sur les fonctions représentables analytiquement. Journal de mathématiques pures et appliquées. **60:** 139–216.
1907 Contribution à l'étude des correspondances de M. Zermelo. Bulletin de la Société Mathématique de France **35:** 202–212.
1907a Sur les transformations ponctuelles, transformant les plans en plans, qu'on peut définir par des procédés analytiques. Accademia delle Scienze di Torino, Classe di Scienze Fisiche, Matematiche, e Naturale (Atti) **42:** 532–539.
1918 Remarques sur les théories de la mesure et de l'intégration. Annales scientifiques de l'Ecole Normale Supérieure (3) **35:** 191–250.
1922 Notice sur les travaux scientifiques. E. Privat. Toulouse.
1941 Les controverses sur la théorie des ensembles et la question des fondements. Les entretiens de Zurich, F. Gonseth, Ed.: 109–122. Leeman. Zurich.
LECHELAS, G.
1897 Review of Couturat, 1896. Revue de Métaphysique et de Morale **5:** 462–488, 620–643.
LEVI, B.
1902 Intorno alla teoria degli aggregati. Istituto Lombardo di Scienze e Lettere, Rendiconti (2) **35:** 863–868.

MOORE, G. H.
1978 The Origins of Zermelo's Axiomatization of Set Theory. Journal of Philosophical Logic **7:** 307–329.
1982 Zermelo's Axiom of Choice: Its Origins, Development, and Influence. Springer. New York.
MORSE, A. P.
1949 Squares are normal. Fundamenta Mathematicae **36:** 35–39.
PEANO, G.
1890 Démonstration de l'intégrabilité des équations différentielles ordinaires. Mathematische Annalen **37:** 182–228.
SHELAH, S.
1980 Going to Canossa. Abstracts of the American Mathematical Society **1:** 630.
SIERPIŃSKI, W.
1916 Sur le rôle de l'axiome de M. Zermelo dans l'analyse moderne. Comptes Rendus Hebdomadaires des Séances de l'Académie des Sciences, Paris **163:** 688–691.
1917 Sur quelques problèmes qui impliquent des fonctions non-mesurables. *ibid.* **164:** 882–884.
1918 L'axiome de M. Zermelo et son rôle dans la théorie des ensembles et l'analyse. Bulletin de l'Académie des Sciences de Cracovie, Classe des Sciences Mathématiques, Série A: 97–152.
1941 L'axiome du choix et l'hypothèse du continu. Les entretiens de Zurich. F. Gonseth, Ed.: 125–143. Leeman. Zurich.
SOLOVAY, R.
1965 The Measure Problem. Notices of the American Mathematical Society **12:** 217.
1970 A Model of Set Theory in which Every Set of Reals Is Lebesgue Measurable. Annals of Mathematics **92:** 1–56.
1971 Real-Valued Measurable Cardinals. Axiomatic set theory. Proc. Symp. Pure Math. (Amer. Math. Soc.) **13**(1): 397–428.
STEPRĀNS, J.
1982 Some Results in Set Theory. Ph.D. dissertation, University of Toronto.
TANNERY, J.
1884 Review of Cantor, 1883b–k. Bulletin des sciences mathématiques et astronomiques (**2**) **2:** 162–171.
1886 Introduction à la théorie des fonctions d'une variable. Hermann. Paris.
1901 Notice sur les travaux scientifiques. Gauthier-Villars. Paris.
TANNERY, P.
1884 Review of Cantor, 1883b–k. Bulletin des sciences mathématiques et astronomiques (**2**) **2:** 162–171.
1885 Le concept scientifique du continu: Zénon d'Elée et Georg Cantor. Revue philosophique de la France et de l'étranger **20:** 385–410.
TARSKI, A.
1930 Une contribution à la théorie de la mesure. Fundamenta Mathematicae **15:** 42–50.
1962 Some Problems and Results Relevant to the Foundations of Set Theory. Logic, Methodology, and Philosophy of Science (Proceedings of the 1960 International Congress). Stanford University Press. Stanford, CA.: 125–135.
TONELLI, L.
1921 Fondamenti di calcolo delle variazioni, vol. 1. Zanichelli. Bologna.

ULAM, S.
1929 Concerning Functions of Sets. Fundamenta Mathematicae **14:** 231–233.
1930 Zur Masstheorie in der allgemeinen Mengenlehre. *ibid.* **16:** 140–150.
1976 Adventures of a Mathematician. Scribner's. New York.
VITALI, G.
1905 Sul problema della misura dei gruppi di punti di una retta. Tio. Gamberini e Parmeggiani. Bologna.
ZERMELO, E.
1904 Beweis, dass jede Menge wohlgeordnet werden kann (Aus einem an Herrn Hilbert gerichteten Briefe). Mathematische Annalen **59:** 514–516.